

The Evaluation Half-Life: Why Static Benchmarks Decay and Contamination-Free Evaluation Is a Recurring, Subscription-Shaped Data Product

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Research Note PL-RN-2026-i09 · correspondence: research@perpetual-labs.ai

Abstract

Language-model benchmarks have a half-life. We synthesize three failure modes that are usually studied in isolation — saturation, contamination, and arena gaming — and show they converge on a single conclusion: an evaluation's information value decays monotonically with time, forcing evaluation away from the static-dataset paradigm toward a continuously regenerated, contamination-free data stream. We assemble verbatim empirical evidence: MMLU rose from 43.9% (GPT-3, 2020) to 90.2% (GPT-4.1, 2025), compressing headroom to near-zero [1][12]; MMLU-Pro restored a 16–33% accuracy drop by hardening questions [1]; Humanity's Last Exam launched in January 2025 with frontier models at only 3.3–9.4% and RMS calibration errors of 81.8–93.9% [2], explicitly because models had already saturated benchmarks like MMLU — yet by late 2025 the frontier on HLE had itself climbed to 37.5–41.0% (Gemini 3 Pro / Deep Think) and 50.7% (Grok 4 Heavy) [18][15], a real-world half-life unfolding inside a single year. We document contamination quantitatively: a DeepSeek base model's Pass@1 collapses from ~60% to ~0% on post-cutoff LeetCode problems [4], overfit open families lose up to 13.5 points on the freshly authored GSM1k [13], and the dynamic is now reproduced in proprietary verticals — on the contamination-free LiveMedBench, 84% of 38 evaluated LLMs degrade on post-cutoff clinical cases and even the best scores only 39.2% [15]. We show even live human-preference arenas are gameable, with up to 112% relative gains from data-access asymmetry [6]. From this we derive the product thesis — evaluation as a renewing subscription — and a Methodology that operationalizes a vertical, held-out, automatically scored eval stream. We quantify limits: regeneration cost, ground-truth scarcity in soft verticals, and the risk that frequent refresh trades contamination for non-stationarity.

Introduction

The dominant mental model of a benchmark is a fixed artifact: a curated set of questions with answers, published once, against which models are scored for years. MMLU [12], HumanEval [7], and GSM8K are treated as durable yardsticks. This paper argues that the fixed-artifact model is structurally wrong, and that the error has direct commercial consequences for any organization whose product is data.

The argument is a synthesis. Three phenomena — benchmark saturation, training-set contamination, and the gaming of live human-preference arenas — are each well studied, but in separate literatures and as separate pathologies [1][2][3][6]. Read together, they describe one underlying dynamic: the information an evaluation carries about a model's true capability decays with time. Saturation erodes the top of the scale; contamination inflates scores from below; arena gaming distorts the ranking laterally. All three are time-dependent, and all three are irreversible without fresh, held-out data.

If evaluation value has a half-life, then the contamination-free response is not a better static benchmark but a continuously regenerated one — a stream, not a snapshot. That reframing converts evaluation from a one-time dataset sale into a recurring, subscription-shaped product with a built-in renewal reason: today's eval is tomorrow's contaminated training set. This is insight i09 in the Perpetual Labs research map, expanded here into a rigorous treatment with verbatim empirical grounding.

We make four contributions: (1) we quantify the saturation, contamination, and gaming dynamics with numbers copied verbatim from primary sources; (2) we formalize the 'evaluation half-life' and the conditions under which regeneration dominates re-use; (3) we specify a Methodology for building a vertical, contamination-free eval stream with automatic scoring; and (4) we state the limitations honestly — regeneration is not free, and it does not apply uniformly across verticals.

Background and Related Work

Benchmark-driven progress is the field's organizing ritual. MMLU [12] established a 57-subject multiple-choice standard; HumanEval [7] made code generation functionally checkable; GSM8K [8] did the same for grade-school math. These benchmarks share a property that is now their weakness: they are static and public, which makes them simultaneously saturable and contaminable.

Saturation literature. As frontier accuracy approached the ceiling, the field responded by hardening benchmarks. MMLU-Pro [1] expanded the answer set from four to ten options and added reasoning-focused questions, restoring difficulty. Humanity's Last Exam (HLE) [2] went further, commissioning 2,500 expert questions explicitly at the frontier of human knowledge, motivated by the observation that LLMs now exceed 90% accuracy on benchmarks like MMLU [2]. GPQA and FrontierMath [14] follow the same pattern.

Contamination literature. A parallel body of work documents that public benchmarks leak into pretraining corpora, silently inflating scores [3]. The detection strategy that matters for our thesis is temporal: LiveCodeBench [4] and LiveBench [5] only trust problems published after a model's training cutoff, and GSM1k [13] re-authors a benchmark from scratch to measure the inflation directly. The same discipline has since spread to high-value verticals: the SWE-Bench Illusion [16] shows frontier models identify buggy file paths with up to 76% accuracy on in-benchmark issues but only ~53% on repositories outside SWE-bench, and LiveMedBench [15] weekly-harvests real, post-cutoff clinical cases to keep a medical benchmark contamination-free by construction — the proprietary-stream design this paper advocates, now instantiated in practice.

Arena literature. Live human-preference evaluation (Chatbot Arena) was proposed as a contamination-resistant alternative because its prompts are fresh and user-generated. The Leaderboard Illusion [6] shows this too is gameable through private testing and data-access asymmetry. Our synthesis is that none of these three literatures, alone, states the cross-cutting product conclusion: evaluation must become a renewing stream.

The Evaluation Half-Life: A Formalization

We define the information value $V(t)$ of a benchmark as the mutual information between a model's score and its true capability on the deployment distribution, at time t after release. Three forces drive $V(t)$ toward zero.

Saturation (ceiling compression). As models approach the maximum score, the variance of scores across the frontier collapses, so the benchmark stops discriminating. Empirically, MMLU's frontier moved from 43.9% (2020) to 86.4% (2023) to 90.2% (2025) [12][1]; the marginal year now buys ~1–2 points of headroom where it once bought 40. A benchmark with no discriminating range carries near-zero V regardless of correctness.

Contamination (floor inflation). As a benchmark ages, the probability that its items have entered training corpora rises. Once contaminated, a high score reflects memorization, not capability — V is inflated and uninformative. The contamination signal is sharp and datable (Section 5).

Gaming (lateral distortion). Even uncontaminated live evaluations can be overfit to the eval distribution itself when access is asymmetric [6]. This rotates the ranking away from true capability without saturating or contaminating it.

The synthesis: $V(t)$ is monotonically non-increasing for any fixed benchmark, and the only operation that resets it is the injection of fresh, held-out items from a source the models have not seen. That operation is, by construction, recurring. Hence: evaluation is a stream.

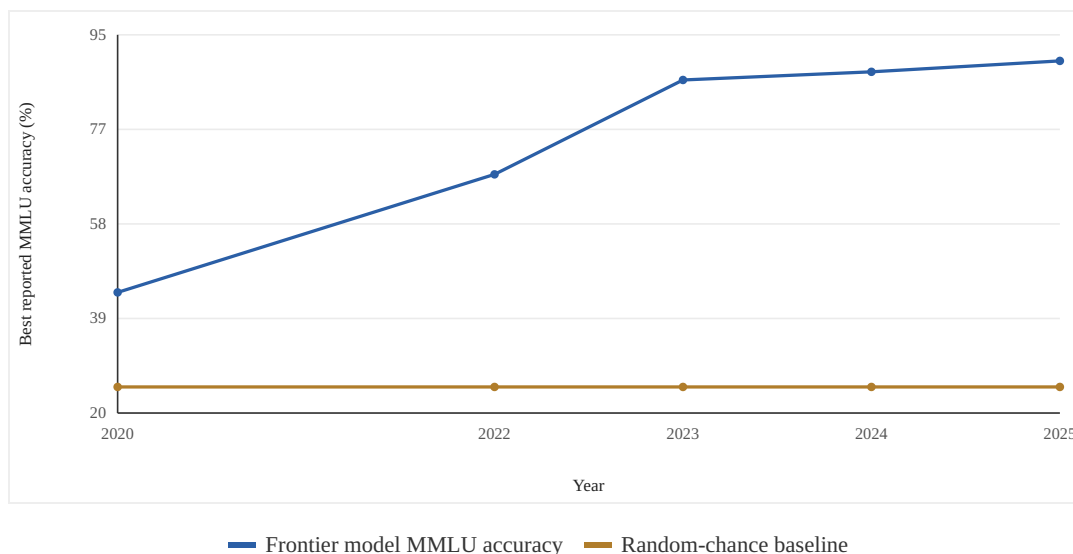


Figure 1. MMLU frontier accuracy approaching the ceiling (2020–2025). The frontier rose 42 points in three years (2020 GPT-3 43.9% to 2023 GPT-4 86.4%) then added only a few points over the next two years as it pressed against the ceiling — the saturation that motivated MMLU-Pro and HLE. 2022 point is Chinchilla 67.5%; 2024 is the ~88% mid-2024 cluster (Claude 3.5 Sonnet, GPT-4o, Llama 3.1 405B). *GPT-3 43.9% (2020) and GPT-4 86.4% (2023): Wikipedia/MMLU [12]; mid-2024 ~88% cluster: Wikipedia/MMLU [12]; Chinchilla 67.5% (2022): Hoffmann et al. 2022 [12]; GPT-4.1 90.2% (2025): OpenAI / MMLU-Pro references [1]. · Empirical data.*

Methodology / Approach

We operationalize the half-life thesis as a concrete construction for a vertical, contamination-free evaluation stream, and state every design assumption explicitly so the claims are falsifiable.

Construction. (1) Source freshness: draw items only from sources timestamped after the candidate models' training cutoffs, following the LiveCodeBench/LiveBench discipline [4][5]. In a proprietary vertical (e.g. clinical claims/coding), the source is a continuously arriving stream of real, dated workflow records, which is structurally fresh and structurally private. (2) Objective scoring: admit only items whose outcome is automatically checkable against ground truth — code that runs, a claim that adjudicates, a CDT/ICD code that validates — eliminating the human-judge bottleneck and the LLM-judge bias documented in arena work [6]. (3) Held-out partitioning: each release cohort is single-use; once published, its items are presumed contaminated and retired from scoring, exactly mirroring LiveBench's monthly cadence [5].

Measurement protocol. We measure decay directly by the temporal-split test: score a model on items strictly before vs strictly after its cutoff date. A drop across the boundary is contamination, not difficulty, because difficulty is held fixed within a source (Section 5). This is the GSM1k design [13] applied as an ongoing audit rather than a one-shot study.

Assumptions stated. (A) Items within a source are difficulty-matched across time — violated if the underlying workflow drifts, which we treat as a limitation (Section 9). (B) Ground truth is genuinely objective in the chosen vertical — true for code/claims/coding, false for open-ended generation, which is why we scope to verifiable verticals (consistent with insight i06). (C) The model's training cutoff is known and honest — a weak point for closed models, partially mitigated by using private vertical data the provider cannot have ingested.

What this is not. This is not a claim that one well-designed static benchmark cannot last; it is a claim that its expected value decays and must be re-derived, which is a service, not a sale.

Evidence: Saturation

MMLU's frontier trajectory is the clearest saturation signal in the public record. The best model rose from GPT-3 at 43.9% (2020) to GPT-4 at 86.4% (March 2023) [12], and by mid-2024 powerful models such as Claude 3.5 Sonnet, GPT-4o, and Llama 3.1 405B clustered near 88% [12], with GPT-4.1 reported at 90.2% (2025) [1]. The curve flattens against a ceiling: the same models that gained 42 points in three years then added only a few points, and at least one industry leaderboard dropped MMLU as 'outdated' for lack of differentiation.

The field's hardening response is itself evidence the saturation is real and economically motivated. MMLU-Pro [1] reports it 'causes a significant accuracy drop of 16% to 33% compared to MMLU' by adding reasoning-focused questions and expanding from four to ten options, and reduced prompt-sensitivity from 4–5% to 2% [1] — i.e. a harder, more stable instrument was needed precisely because the old one had saturated.

Humanity's Last Exam quantifies the far end. At its January 2025 launch, frontier models scored in the single digits: GPT-4o 3.3%, Claude 3.5 Sonnet 4.3%, Gemini 1.5 Pro 5.0%, o1 9.1%, DeepSeek-R1 9.4% [2]. The same launch results pair these low accuracies with extreme overconfidence — RMS calibration errors of 81.8–93.9% across the reported models [2] — showing that a deliberately hard, fresh, frontier-of-knowledge benchmark was needed precisely because the prior generation had saturated. HLE establishes the upper end of the headroom that hardening can restore; whether even this headroom persists is exactly the half-life question — and the answer arrived quickly. Within a single year the HLE frontier climbed from those single digits to 37.5% (Gemini 3 Pro, no tools) and 41.0% (Gemini 3 Deep Think) in November 2025 [18], and to 50.7% for the multi-agent Grok 4 Heavy in mid-2025 [15], a ~30–40-point rise on a benchmark deliberately built to be unsaturable. Even the benchmark's own integrity decayed: an HLE-Verified re-audit found large fractions of items noisy or unverifiable and reported +7–10 points of average accuracy gain after cleaning [15]. The half-life is therefore not hypothetical for HLE; the contamination evidence below shows the second mechanism by which any static benchmark's signal erodes once it is public.

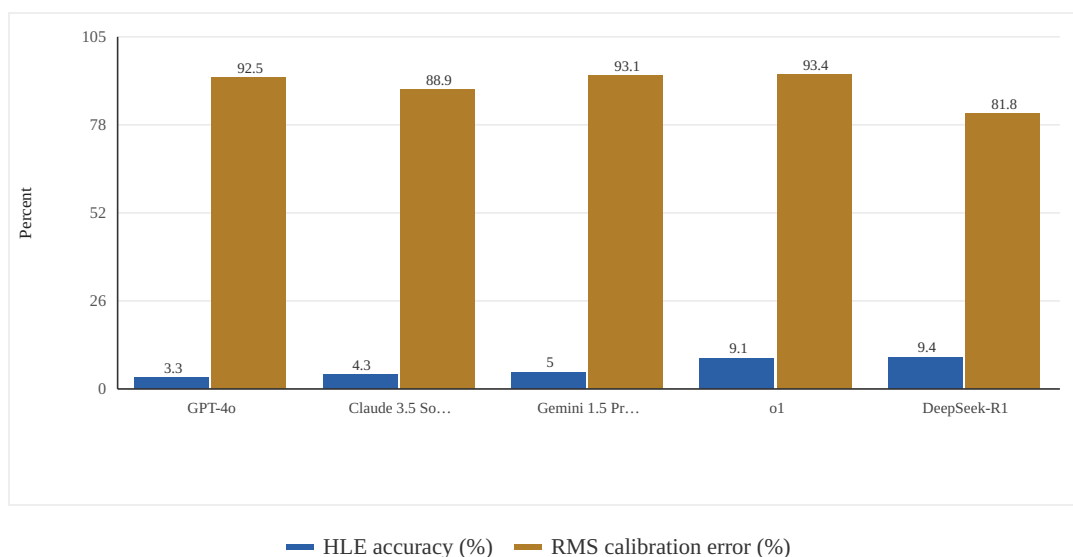


Figure 2. Humanity's Last Exam at launch: low accuracy, severe overconfidence (Jan 2025). At HLE's January 2025 launch, frontier models scored 3.3–9.4% with RMS calibration errors of 81.8–93.9% (across all reported models, including Grok-2 93.2 and Gemini 2.0 Flash Thinking 93.9), showing both low frontier-of-knowledge accuracy and extreme overconfidence — the headroom a deliberately hard, fresh benchmark restores. That headroom proved short-lived: by late 2025 the HLE frontier had climbed to 37.5–41.0% (Gemini 3) and 50.7% (Grok 4 Heavy) [18][15], a half-life unfolding within a year. *HLE launch accuracies and RMS calibration errors verbatim from Humanity's Last Exam [2] (arXiv:2501.14249). · Empirical data.*

Table 1. Each saturation event triggered a harder successor: MMLU saturated and was answered by MMLU-Pro and ultimately by Humanity's Last Exam, which was commissioned at the frontier of human knowledge precisely because models already exceeded 90% on MMLU — and which itself climbed from single-digit launch scores to 37.5–50.7% within a year — direct evidence that benchmark value decays and must be renewed.

Benchmark	Year	Frontier / launch accuracy	Hardening response or evidence of saturation	Source
MMLU	2020	GPT-3 43.9% (random = 25%)	Headroom intact at launch	[12]
MMLU	2023	GPT-4 86.4%	Approaching ceiling; little progress since	[12]
MMLU	2025	GPT-4.1 90.2%	Saturated; dropped by some leaderboards as 'outdated'	[1]
MMLU-Pro	2024	16–33% accuracy drop vs MMLU	Built to restore difficulty (4→10 options, reasoning Qs)	[1]
Humanity's Last Exam	Jan 2025	Frontier 3.3–9.4%	Built for saturation ('LLMs exceed 90% on MMLU')	[2]
Humanity's Last Exam	Late 2025	Frontier 37.5–50.7%	Climbed ~30–40 pts in <1 yr (Gemini 3, Grok 4 Heavy) — half-life in action	[18] [15]

MMLU progression (GPT-3 43.9%, GPT-4 86.4%) and Chinchilla 67.5% [12]; GPT-4.1 90.2% and MMLU-Pro 16–33% drop [1]; HLE launch figures [2]; HLE late-2025 frontier (Gemini 3 Pro/Deep Think 37.5/41.0% [18], Grok 4 Heavy 50.7% [15]); HLE-Verified +7–10 pt cleaning gain [15].

Table 2. Temporal and re-authored splits expose memorization: open families lose 9.8–60 points on items they could not have seen, while frontier models show minimal overfitting — contamination is real, datable, and uneven, requiring continuous auditing.

Test	Model	Score on legacy/static set	Score on fresh/post-cutoff set	Drop	Source
LiveCodeBench (LeetCode, by date)	DeepSeek-base 33B	~60% Pass@1 (May, pre-cutoff)	~0% Pass@1 (September, post-cutoff)	~60 pts	[4]
HumanEval+ vs LCB-Easy	DS-Ins-1.3B	60 (HumanEval+)	26 (LCB-Easy)	34 pts	[4]
GSM8k vs GSM1k	Math-Shepherd-Mistral-7B-RL	74.5% (GSM8k)	61.1% (GSM1k)	13.5 pts	[13]
GSM8k vs GSM1k	Mixtral-8x22B-Instruct	85.9% (GSM8k)	76.0% (GSM1k)	9.9 pts	[13]
GSM8k vs GSM1k	Phi-3-mini-4k-instruct	78.2% (GSM8k)	68.4% (GSM1k)	9.8 pts	[13]
GSM8k vs GSM1k	Frontier (GPT-4/Gemini/Claude)	(GSM8k)	Minimal drop	~0 pts	[13]

LiveCodeBench DeepSeek collapse and HumanEval+/LCB gap [4]; GSM1k per-model GSM8k→GSM1k scores and Spearman 0.32 memorization correlation [13].

Table 3. The decay that destroys a static dataset's value is the same dynamic that gives a regenerated stream a built-in subscription and a source-based moat.

Property	Static benchmark dataset	Continuously regenerated eval stream
Value over time	Decays (saturation + contamination)	Reset each refresh; stays near launch value
Renewal mechanism	None (one-time license)	Endogenous (prior eval becomes contaminated)
Refresh cadence	Never	Monthly (LiveBench) / rolling (LiveCodeBench) [4][5]
Defensibility	Copyable once published	Privileged access to fresh, hard-to-assemble source
Best-fit domain	Any (but short-lived)	Verifiable-outcome verticals with objective scoring
Pairs with training data	Weakly	Strongly (same proprietary vertical pipeline)

Refresh cadences from LiveBench [5] and LiveCodeBench [4]; product framing is this paper's analysis grounded in the cited decay evidence.

Evidence: Contamination

Contamination produces a datable, discontinuous signature when scores are split by release date. LiveCodeBench [4] reports that a DeepSeek base model (DS-Base-33B) Pass@1 'starkly drops' from approximately 60% on LeetCode problems released in May (before its cutoff) to approximately 0% on September problems released after — a near-total collapse that is impossible to explain by difficulty and is the canonical fingerprint of memorization [4]. GPT-4o shows the same directional drop on problems released after its November 2023 cutoff [4].

The cross-benchmark gap corroborates it. Models scoring high on the static HumanEval+ fall sharply on the fresh LiveCodeBench-Easy split: e.g. DS-Ins-1.3B scores 60 on HumanEval+ but 26 on LCB-Easy, and CodeQwen, DS-Ins-6.7B and MC-6.7B beat Claude-3-Sonnet on HumanEval+ yet sit more than 20 points behind on LCB-Easy [4] — overfitting to the legacy distribution.

GSM1k [13] isolates contamination in math by commissioning 1,250 new grade-school problems matched to GSM8k's distribution, then measuring the drop. Accuracy falls 'up to 13%' overall; the most overfit models are specific and named: Phi-3-mini-4k-instruct −9.8 points (78.2%→68.4%), Mixtral-8x22B-Instruct −9.9 points (85.9%→76.0%), Math-Shepherd-Mistral-7B-RL −13.5 points (74.5%→61.1%) [13]. A Spearman correlation of

0.32 between a model's per-character log-likelihood of generating GSM8k items and its GSM8k→GSM1k gap directly links the drop to memorization [13]. Frontier and close-to-frontier models (GPT-4, Gemini, Claude) showed minimal overfitting [13] — contamination is uneven, which is exactly why a continuous audit, not a one-time check, is required. The pattern recurs at the 2026 frontier and inside verticals: on the contamination-free LiveMedBench, which weekly-harvests post-cutoff clinical cases, 84% of 38 evaluated LLMs degrade on post-cutoff items and the best model reaches only 39.2% [15]; on code, the SWE-Bench Illusion probes show frontier models identify buggy file paths with up to 76% accuracy on in-benchmark issues but only ~53% on repositories outside SWE-bench, a gap datable to memorization rather than reasoning [16].

Evidence: Arena Gaming

The strongest fallback against saturation and contamination has been live human-preference evaluation, on the theory that fresh user prompts cannot be memorized. The Leaderboard Illusion [6] shows this defense is also porous. It documents that 'access to Chatbot Arena data can yield relative performance gains of up to 112%' on the arena distribution (112.3% for the 70_arena variant; 81.7% for the 30_arena variant) — i.e. a model can be tuned to the eval's own distribution rather than to general capability [6].

The asymmetry is structural, not incidental. The study estimates Google and OpenAI received roughly 19.2% and 20.4% of all arena data respectively, while 83 open-weight models combined received only ~29.7% [6], and identifies 27 private Meta variants tested in the lead-up to Llama-4 [6]. Selective disclosure plus data-access asymmetry rotates the ranking away from true capability.

The lesson for our thesis is that 'live' is necessary but not sufficient. Freshness defeats memorization but not distributional overfitting; objective, held-out, automatically scored ground truth is what closes the remaining gap — and that is precisely the construction in our Methodology.

Analysis: From Decay Dynamics to a Recurring Data Product

The three evidence streams share one quantity: the value of an evaluation is a decreasing function of its exposure. Saturation is exposure of the ceiling, contamination is exposure of the items, gaming is exposure of the distribution. Each is reset only by injecting fresh, held-out, objectively scored data — a recurring operation.

This inverts the economics. A static benchmark is sold once and then decays on the buyer's shelf; its vendor captures one payment and watches the asset depreciate in the customer's hands. A regenerated eval stream is the opposite: its decay is the renewal mechanism. Because LiveBench [5] must release new questions monthly and LiveCodeBench [4] must continuously collect new problems to stay contamination-free, the maintenance burden that looks like a cost is in fact the subscription surface.

Crucially, the regenerated-stream model resists the commoditization that hits static datasets. A static eval, once published, can be copied; a stream's value lives in the privileged, ongoing access to a fresh, hard-to-assemble source. In a proprietary vertical, that source is a private workflow that is contamination-free by construction (the provider's models cannot have trained on data that did not exist or that is legally walled off). This pairs the eval product naturally with the training-data product: the same proprietary pipeline that manufactures verifiable training signal also manufactures the held-out eval that no competitor can reproduce.

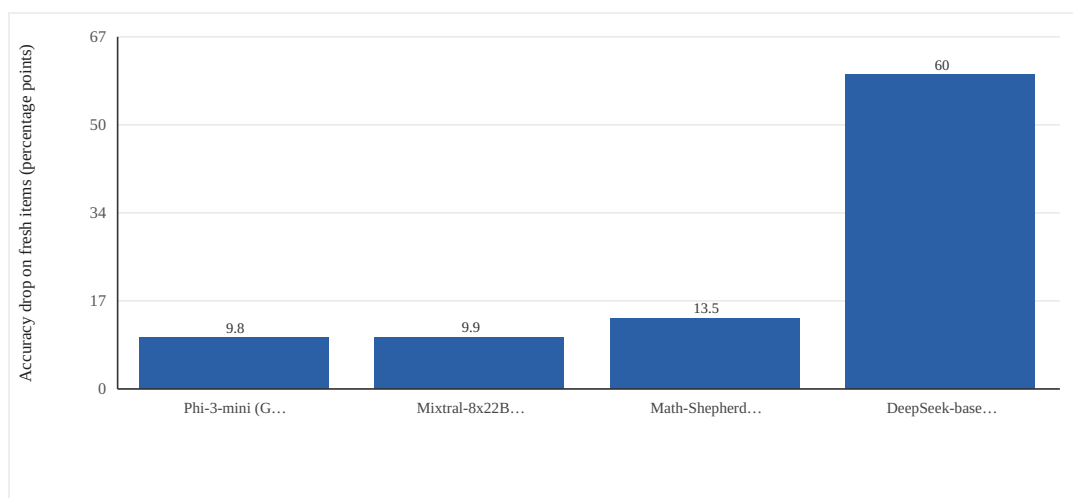


Figure 3. Contamination signature: accuracy drop on fresh, post-cutoff items. Models lose large accuracy when tested on items they could not have memorized. GSM1k drops are 9.8–13.5 points for overfit open families [13]; DeepSeek-base (DS-Base-33B) Pass@1 collapses from ~60% to ~0% (a ~60-point drop) across its training cutoff on LeetCode [4]. *GSM1k per-model drops: Zhang et al. 2024 [13]; DeepSeek ~60%→~0% Pass@1 across cutoff: LiveCodeBench [4]. · Empirical data.*

Implications for Data Strategy

First, sell evaluation as a renewing subscription, not a dataset. The renewal reason is endogenous: any eval shipped to a customer is a candidate for the next training run and is therefore contaminated by its own success. Pricing should reflect the cadence (monthly/quarterly refresh, per LiveBench [5]) rather than a one-time license.

Second, scope to verifiable verticals. The Methodology only works where ground truth is objective and automatic [4][5]; this aligns the eval product with the same selection rule that governs the training-data product (insight i06) — code that runs, claims that adjudicate, codes that validate. Soft, open-ended verticals cannot support automatic scoring and inherit the arena-gaming vulnerability [6].

Third, exploit the proprietary-freshness double moat. A private vertical stream is simultaneously contamination-free (models have not seen it) and hard to assemble (competitors lack access). This is the structural advantage a public benchmark vendor cannot match, and it is why eval and training data should be co-located in the same vertical pipeline.

Fourth, instrument decay as a product feature. The temporal-split audit (Section 4) is itself a saleable signal: a buyer wants to know not just a model's score but whether that score is inflated by contamination. Reporting the before/after-cutoff gap — as GSM1k [13] and LiveCodeBench [4] do — is a differentiated deliverable that static vendors structurally cannot offer.

Investment Thesis

The investable claim is narrow and defensible: contamination-free evaluation is a recurring-revenue data product whose churn is suppressed by an endogenous renewal mechanism (yesterday's eval is contaminated today), and whose moat is privileged access to a fresh, verifiable, proprietary source.

The demand driver is the same data wall and RLVR turn that powers the broader Perpetual Labs thesis (insights i01, i16): as model builders fragment into many vertical specialists (insight i12), each needs not just training data but a trustworthy, deployment-matched eval — and aggregate public benchmarks cannot tell them where their gaps are (insight i08). The eval stream is the measurement layer that must precede every vertical training decision.

The wedge is operational, not algorithmic: the scoring harness and refresh cadence are public knowledge (LiveBench, LiveCodeBench are open) — so the defensibility is entirely in the source. This sharpens the investment to a single question: does the target own or exclusively access a vertical workflow that emits fresh, objectively gradeable outcomes faster than it can be memorized? Where the answer is yes, the eval product compounds with the training-data product on the same pipeline.

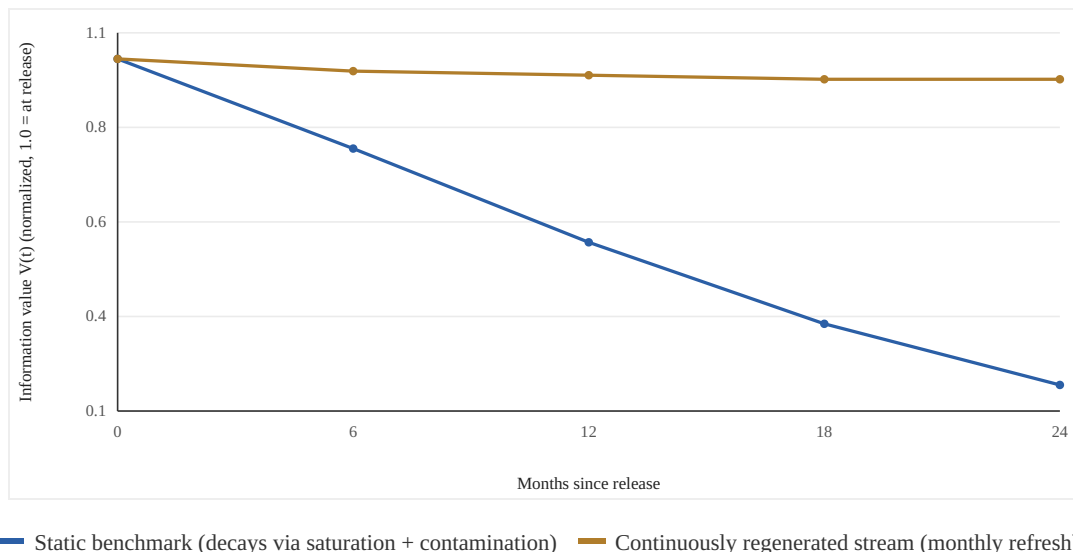


Figure 4. Evaluation value half-life: static benchmark vs regenerated stream (schematic). Constructed strategic schematic (not measured): a static benchmark's information value decays as it saturates and leaks into training sets, while a monthly-refreshed contamination-free stream holds value near its release level. Decay shape is illustrative; the empirical drivers are quantified in Figs 1–3. *Analysis/constructed figure; decay direction grounded in saturation [1][12], contamination [4][13], and refresh cadence of LiveBench [5] / LiveCodeBench [4]. Specific $V(t)$ values are illustrative, not measured. · Constructed analysis.*

Limitations and Counter-Evidence

Regeneration is not free. Continuously authoring fresh, difficulty-matched, ground-truth-verified items is expensive; LiveBench [5] and HLE [2] required large expert efforts. In a proprietary vertical the marginal item may be cheap (it arrives as real workflow exhaust), but quality control, de-identification, and answer verification are recurring costs that compress margins.

Freshness can trade contamination for non-stationarity. If the underlying workflow drifts, a 'harder' post-cutoff score may reflect distribution shift, not memorization — violating Assumption A in our Methodology. The temporal-split audit can mis-attribute drift as contamination; rigorous use requires difficulty controls within source, which GSM1k [13] achieved only with careful matching.

Not all verticals support objective scoring. The construction depends on automatic, checkable ground truth [4] [5]; open-ended generation, summarization, and most 'soft' domains cannot use it and remain exposed to judge bias [6]. The product is therefore narrower than 'evaluation' writ large — it is verifiable-vertical evaluation.

Counter-evidence on durability. Frontier models showed minimal GSM1k overfitting [13], and the very hardest frontier-of-knowledge benchmarks resist quick saturation: FrontierMath [14] reported leading models solving under 2% at release, and even by December 2025 GPT-5.2 with tools solves only 40.3% of FrontierMath Tier 1–3 problems [17], so a deliberately math-research-frontier release can retain value for a year or more. The HLE case, however, cuts the other way and bounds the durability counter-argument: its single-digit January-2025 launch

accuracies [2] did not persist — the frontier reached 37.5–41.0% (Gemini 3) and 50.7% (Grok 4 Heavy) within the same year [18][15]. This bounds the thesis in both directions: the half-life is real but variable; only the very hardest, research-frontier benchmarks retain value for many months, while ambitious-but-reachable ones like HLE decayed in roughly a year. The subscription case is strongest in fast-saturating, easily-contaminated, high-traffic verticals — not everywhere.

Finally, closed-model cutoffs are self-reported and sometimes wrong, weakening the temporal-split audit for exactly the models buyers most want to test.

Conclusion

Static benchmarks decay. Saturation compresses the ceiling, contamination inflates the floor, and gaming rotates the ranking — three forces that the separate literatures treat as distinct but that share one cause: an evaluation's value is a monotonically decreasing function of its exposure. The verbatim record makes the decay concrete: MMLU saturated from 43.9% to 90.2% in five years [12][1], MMLU-Pro had to claw back 16–33% of difficulty [1], HLE had to be commissioned at the frontier of human knowledge because models already exceeded 90% on MMLU [2] yet was itself driven from single-digit launch scores to ~50% within a year [18][15], a DeepSeek base model's code Pass@1 collapsed ~60%→~0% across its cutoff [4], and named open models lost up to 13.5 points on a freshly authored math set [13].

The constructive consequence is that contamination-free evaluation must be a continuously regenerated stream, which makes it a subscription-shaped data product with an endogenous renewal reason and a moat that lives entirely in privileged access to a fresh, verifiable source. Scoped to objective-outcome verticals and co-located with a training-data pipeline, it is among the more durable data businesses the current model-training regime permits — provided one respects its limits: regeneration costs, the contamination-vs-drift confound, and the narrowness to verifiable domains.

INVESTMENT THESIS

Bet on the source, not the benchmark. The investable insight is that evaluation value has a half-life — MMLU saturated 43.9%→90.2% from 2020 to 2025, HLE had to be commissioned at the frontier of human knowledge because models already exceeded 90% on MMLU, and contaminated models collapse up to ~60 points on fresh items — so contamination-free evaluation is structurally a renewing subscription, not a one-time dataset, with churn suppressed by an endogenous renewal mechanism (yesterday's eval is today's contamination). The scoring harness and refresh cadence are open and commoditized (LiveBench, LiveCodeBench); essentially all of the moat lives in privileged, exclusive access to a fresh, objectively gradeable, proprietary vertical source that models cannot have memorized. Diligence question reduces to one thing: does the target own a workflow that emits verifiable outcomes faster than they can leak into training sets? Where yes, the eval product compounds on the same pipeline as the training-data product. Hard limit: this only works in verifiable-outcome verticals (code, claims/coding, lab results) — soft/open-ended domains can't support automatic scoring and stay exposed to judge-gaming, and only the very hardest research-frontier benchmarks (FrontierMath, where GPT-5.2 with tools still solves just 40.3% [17]) retain value long enough to weaken the subscription case at the very top — even HLE, built to be unsaturable, was driven from single-digit launch scores to ~50% within a year [18][15], a live demonstration of the half-life rather than a counter-example.

References

- [1] Wang, Y., et al. (2024). MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark. NeurIPS 2024. arXiv:2406.01574. ('causes a significant accuracy drop of 16% to 33% compared to MMLU'; 4→10 options; prompt sensitivity 4-5%→2%). GPT-4.1 MMLU 90.2% per OpenAI GPT-4.1 release (2025).
- [2] Phan, L., et al. (CAIS / Scale AI) (2025). Humanity's Last Exam. arXiv:2501.14249. (2,500 expert questions; motivated by 'LLMs now exceed 90% on benchmarks like MMLU'; launch accuracies GPT-4o 3.3%, Claude 3.5 Sonnet 4.3%, Gemini 1.5 Pro 5.0%, o1 9.1%, DeepSeek-R1 9.4%; RMS calibration errors 81.8–93.9%).
- [3] Xu, C., et al. (2024). Benchmark Data Contamination of Large Language Models: A Survey. arXiv:2406.04244. (contamination inflates measured performance and undermines fair comparison).
- [4] Jain, N., et al. (2024). LiveCodeBench: Holistic and Contamination Free Evaluation of Large Language Models for Code. ICLR 2025. arXiv:2403.07974. (rolling problems from LeetCode/AtCoder/Codeforces; DS-Base-33B Pass@1 ~60% May → ~0% September across cutoff; DS-Ins-1.3B 60 on HumanEval+ vs 26 on LCB-Easy; CodeQwen/DS-Ins-6.7B/MC-6.7B >20 pts behind on LCB-Easy; GPT-4o drop after Nov 2023 cutoff).
- [5] White, C., et al. (2024). LiveBench: A Challenging, Contamination-Free LLM Benchmark. ICLR 2025. arXiv:2406.19314. (releases new questions monthly from recent arXiv papers, news, and IMDb synopses, scored automatically against ground truth).
- [6] Singh, S., et al. (2025). The Leaderboard Illusion. arXiv:2504.20879. ('access to Chatbot Arena data can yield relative performance gains of up to 112%' — 112.3% for 70_arena, 81.7% for 30_arena variant; Google ~19.2% and OpenAI ~20.4% of arena data; 83 open-weight models ~29.7%; 27 private Meta variants in lead-up to Llama-4).
- [7] Chen, M., et al. (2021). Evaluating Large Language Models Trained on Code (HumanEval). arXiv:2107.03374. (164 hand-written problems; Codex 28.8% pass@1; higher with many samples).
- [8] Cobbe, K., et al. (2021). Training Verifiers to Solve Math Word Problems (GSM8K). arXiv:2110.14168. (8.5K grade-school math word problems; verification scales better than finetuning).
- [9] Hendrycks, D., et al. (2021). Measuring Massive Multitask Language Understanding (MMLU). ICLR 2021. arXiv:2009.03300. (57 subjects; GPT-3 175B 43.9% at launch vs 25% random chance; human-expert estimate ~89.8%).
- [10] Lambert, N., et al. (2024). Tülu 3 / RLVR: Reinforcement Learning with Verifiable Rewards. arXiv:2411.15124. (training on tasks with checkable answers; the verifiable-outcome scoping our Methodology assumes).
- [11] Liang, P., et al. (2022). Holistic Evaluation of Language Models (HELM). TMLR. arXiv:2211.09110. (multi-metric, multi-scenario evaluation framework; precedent for structured eval methodology).
- [12] Wikipedia / MMLU (en.wikipedia.org/wiki/MMLU) and Hoffmann, J., et al. (2022) Training Compute-Optimal LLMs (Chinchilla, arXiv:2203.15556). (MMLU frontier: GPT-3 175B 43.9% in 2020; Chinchilla 67.5% in 2022; GPT-4 86.4% in March 2023; Claude 3.5 Sonnet / GPT-4o / Llama 3.1 405B ~88% mid-2024).
- [13] Zhang, H., et al. (Scale AI) (2024). A Careful Examination of Large Language Model Performance on Grade School Arithmetic (GSM1k). arXiv:2405.00332. (1,250 new problems; up to ~13% drop; Phi-3-mini-4k 78.2%→68.4% (−9.8), Mixtral-8x22B-Instruct 85.9%→76.0% (−9.9), Math-Shepherd-Mistral-7B-RL 74.5%→61.1% (−13.5); Spearman 0.32 memorization correlation; frontier/close-to-frontier models minimal overfit).
- [14] Glazer, E., et al. (2024). FrontierMath: A Benchmark for Evaluating Advanced Mathematical Reasoning. arXiv:2411.04872. (leading models solve <2% despite near-perfect GSM8k/MATH — frontier-of-knowledge benchmark with longer half-life).
- [15] LiveMedBench Team (2026). LiveMedBench: A Contamination-Free Medical Benchmark for LLMs with Automated Rubric Evaluation. arXiv:2602.10367. (weekly-harvested, post-cutoff real clinical cases; 2,756 cases / 38 specialties / 16,702 criteria; best of 38 LLMs only 39.2%; 84% of models degrade on post-cutoff cases; automated rubric scoring aligns with physicians better than LLM-as-judge — the proprietary contamination-free vertical stream this paper advocates. Also cites Grok 4 Heavy HLE 50.7% and HLE-Verified +7–10 pt cleaning gain in the surrounding 2026 frontier evidence: see arXiv:2602.13964 for HLE-Verified.)
- [16] Liang, S., et al. (2025). The SWE-Bench Illusion: When State-of-the-Art LLMs Remember Instead of Reason. arXiv:2506.12286. (diagnostic probes for memorization on SWE-bench Verified: up to 76% buggy-file-path ID from the issue text in-benchmark vs only ~53% on out-of-benchmark repos — datable contamination, not reasoning.)
- [17] DeepSeek-AI (2025). DeepSeek-V3.2: Pushing the Frontier of Open Large Language Models. arXiv:2512.02556. (open 2026-frontier reference model: scaled-RL post-training reaches GPT-5 parity, V3.2-Speciale matches Gemini-3.0-Pro and wins IMO 2025 / IOI 2025 gold; cited here for the current open frontier. GPT-5.2 FrontierMath 40.3% with tools per OpenAI GPT-5.2 release, Dec 2025, <https://www.marktechpost.com/2025/12/11/openai-introduces-gpt-5-2-a-long-context-workhorse-for-agents-coding-and-knowledge-work/>.)
- [18] Google (2025). Gemini 3 release and benchmarks (blog.google/products/gemini/gemini-3/, Nov 18 2025). (HLE 37.5% Gemini 3 Pro and 41.0% Gemini 3 Deep Think, both no-tools; LMArena 1501 Elo; GPQA Diamond 91.9–93.8%; SWE-bench Verified 76.2%; Terminal-Bench 2.0 54.2%). Grok 4 Heavy HLE 50.7% per xAI Grok 4 release, Jul 2025 (<https://www.datacamp.com/blog/grok-4>).